NILAKSHI ROY
Data Scientist
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Portfolio: roynilakshi91.github.io/Nilakshi_Roy_portfolio

WORK EXPERIENCE

TikTok USDS, Data Science and Analytics, San Jose, CA Aug 2022 – Present

- Conducted exploratory data analysis across large-scale datasets, developed end-to-end data pipelines, and built **20+ interactive dashboards** with real-time alerts for metric monitoring and data quality, while collaborating with cross-functional stakeholders.
- Built **predictive machine learning models (CatBoost)** to mitigate abnormal fund flow risk and improve user experience, **reducing appeals from 1,400 to 690 per day** and lowering the **chargeback rate from 4% to 1.6%**.
- Systematized pre-launch performance evaluations using **A/B testing** experimental design and champion-challenger frameworks to optimize decision thresholds and minimize **false-positive rates** to less than **0.01%**.
- Built agentic AI workflows using **Claude and MCP** to automate recurring, multi-step operational tasks- data retrieval, weekly performance analysis, metric monitoring and reporting, **reducing manual effort by 86%, and** saving **15+ hours per week.**
- Developed Trust Layer ML solutions using **TF-IDF text vectorization with gradient boosting** to generate confidence scores, ensuring VIP user experience and driving **$11K/day in incremental revenue**.
- Supported U.S. campaign initiatives and conducted cohort analyses, **causal inference**, and experimentation; applied an ML model to identify coordinated activity, trend manipulation, and inauthentic engagement, achieving a **94% campaign success rate** and **20% increase in user participation**.
- **Prevented $1.7M in capital losses** by developing perception models leveraging **LLMs** to identify abnormal content and detect coordinated cross-regional activity.
- Architected an automated fake-creator detection and enforcement pipeline using registration, content, and behavioral signals, Integrating LLMs model to identify and ban **more than 300 fraudulent accounts** and **12K fake registrations per week**, while maintaining nearly 100% precision.

Google, Data Specialist, Mountain View, CA *Contract* May 2021 – Aug 2022

- Optimized reporting dashboards and planning tools, **reducing data processing time by 80%** and implementing automation through **ETL scripts** to cut weekly manual labor.
- Managed data integrity and account mapping within cloud infrastructure platforms like Salesforce; developed **SQL** test scripts to automate EDA and KPI analysis.
- Translated complex datasets into actionable insights for the GCP Sales operations team, supporting GTM strategy and success through advanced data modeling and creating weekly business reports (WBRs) to ensure systematized data delivery and performance tracking.

Apple Inc., Data Reporting and Testing Analyst, Sunnyvale, CA *Contract* May 2020 – Mar 2021

- Analyzed pre- and post-release reports for iOS systems to identify performance inefficiencies and support risk mitigation efforts for global product launches.
- Leveraged **SQL** to extract and analyze server-side data, developed data-modeling flowcharts for key dimensions and product performance metrics, and built **KPI dashboards** and daily performance reports to communicate insights and support cross-functional stand-ups.

American International Group (AIG), Data Analyst, Bengaluru, India Sep 2016 – Apr 2019

- Analyzed large-scale datasets within financial services to identify patterns and calculate potential capital losses, utilizing **SQL**, Python, and Linux environments.
- Operationalized **Tableau dashboards** to visualize complex risk patterns, translating data findings for stakeholders to support banking and insurance risk mitigation.

SKILLS

- **Programming & Data:** Python (Pandas, NumPy, scikit-learn), SQL, Hive, ETL Pipelines, Data Modeling
- **Machine Learning:** End-to-End ML Pipeline Development, Supervised Learning, CatBoost, XGBoost, TF-IDF Vectorization, Model Evaluation
- **Analytics & Experimentation:** AB Testing, Champion-Challenger Framework, Cohort Analysis, Causal Inference, North Star & Guardrail Metrics, Fraud & Risk Analytics, EDA
- **AI & Automation:** Claude, Model Context Protocol (MCP), Agentic AI Workflows
- **Visualization & Reporting:** Tableau, Power BI, Matplotlib, Seaborn, Microsoft Excel

EDUCATION

- Master of Science in Computer Science - **California State University, East Bay**, Hayward, CA Feb 2020 – Dec 2021
- Master of Technology in Computer Science - **Christ University**, Bengaluru, India Aug 2014 – Jan 2016

ACCOMPLISHMENTS

- **Employee Recognition Award:** Top-quality analytics deliverables at AIG Analytics and Services.
- **Top Performer Award:** Exceeded goals at TikTok, improving detection accuracy and user experience.